

intersection of 2 conics:- $\begin{cases} x^T(v_1 + \mu v_2)x + 2(u_1 + \mu u_2)^T x + f_1 + \mu f_2 = 0 \end{cases}$

$$\Rightarrow \begin{vmatrix} v_1 + \mu v_2 & u_1 + \mu u_2 \\ (u_1 + \mu u_2)^T & f_1 + \mu f_2 \end{vmatrix} = 0$$

$$x^2 + 0y^2 + 0x + y + 0 = 0$$

$$x^2 + 0y^2 + 2x + y + 1 = 0$$

$$v_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; \quad v_2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$u_1^T = \begin{bmatrix} 0 & -1 \end{bmatrix}; \quad u_2^T = \begin{bmatrix} 2 & 1 \end{bmatrix}$$

$$f_1 = 0;$$

$$f_2 = 1$$

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x^2$$

$$\begin{bmatrix} 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = -y$$

$$\begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 2x + y$$

$$\begin{bmatrix} 0 & 2\mu \\ -1 & \mu \end{bmatrix}$$

$$\Rightarrow \begin{vmatrix} \begin{bmatrix} 1+\mu & 0 \\ 0 & 0 \end{bmatrix} & \begin{bmatrix} 2\mu \\ \mu-1 \end{bmatrix} \\ \begin{bmatrix} 2\mu & \mu-1 \end{bmatrix} & \mu \end{vmatrix} = 0$$

$$\begin{vmatrix} 1+\mu & 0 & 2\mu \\ 0 & 0 & \mu-1 \\ 2\mu & \mu-1 & \mu \end{vmatrix} = 0$$

$$\begin{vmatrix} 1+\mu & 0 & 2\mu \\ 0 & 0 & \mu-1 \\ 2\mu & \mu-1 & \mu \end{vmatrix} = 0 :$$

9.1.4.1

$$(\mu-1) \left[(1+\mu)(\mu-1) \right] = 0$$

$$(\mu+1)(\mu-1)^2 = 0.$$

$$\therefore \mu = -1; \mu = 1.$$

Eq 9.1.4.1 $\Rightarrow C_1 + \mu C_2 \rightarrow$ degenerate conics.

$$\Rightarrow (x^2 - y) + \mu(x^2 + 2x + y + 1) = 0$$

Case 1: $\mu = 1;$

$$2x^2 + 2x + 1 = 0 \quad \text{--- (3)}$$

Case 2: $\mu = -1$

$$-2x - 2y - 1 = 0; \quad \text{--- (4)}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \mu \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1+\mu & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$x^T \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + 1$$

$$+ 2x^2 + 2x + 1$$

and

$$\begin{bmatrix} 0 \\ -1 \end{bmatrix} + \mu \begin{bmatrix} 2 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$x^T \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} -2 \\ -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + 1 \Rightarrow 1$$

$$\Rightarrow 2x + 2y + 1 = 0$$

From (3) :

$$2x^2 + 2x + 1 = 0 \rightarrow \text{degenerate conic};$$

$$\text{discriminant: } 4 - 4(2) \Rightarrow -4 < 0$$

No real solutions; $\Rightarrow$ imaginary lines;

i.e.; two conics don't intersect.